

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -1

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Recap of Previous Lecture

Doubt?



Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers

Topics to be Covered



①. 0. 0. 0. ⇒



Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

Topic

SJF Scheduling

OS
P1
P2
P3

→ long term scheduler controls degree of multiprogramming.

→  maintains a good mix of
I/O bound and CPU bound processes.

[MCQ]

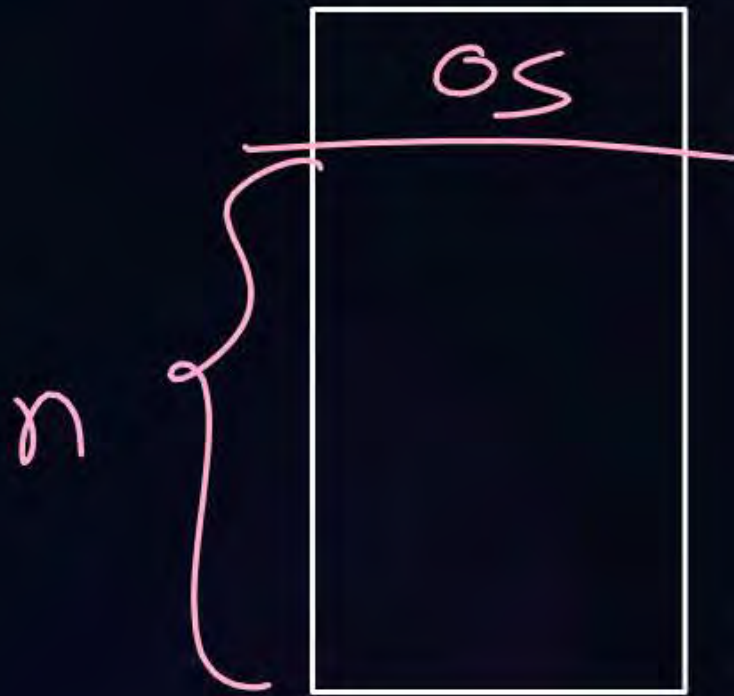
#Q. Which of the following scheduler reduces the degree of multiprogramming?

- A** Short-Term $\Rightarrow$ does not change D.O.M.
- B** Long-Term $\Rightarrow$ can increase —||—
- C** ✓ Mid-Term $\Rightarrow$ can increase and also can decrease D.O.M.
- D** Long-Term and Mid-Term both

[NAT]

#Q. Consider a system with n processes and m CPUs. Maximum and Minimum number of processes in each of the following states possible?

1. Ready state	0	n
2. Running State	0	m
3. Blocked State	0	n
	min	max



$n > m$
↑



Topic : CPU Scheduling



Function:

- Make a selection of one of the ready processes to run next on CPU.

Goal

- Minimize Wait time and Turn-around time
- Maximize CPU utilization (Throughput)
- Fairness





Topic : CPU Scheduling Types

Non-preemptive

Preemptive

Consider processes do not have any I/O requirement.





Topic : Scheduling Times

- Arrival Time (AT): Time at which process arrives
- Burst/Service Time (BT): Amount of time needed for process to run on CPU.
- Completion Time (CT): Time at which process completes execution
- Turn-Around Time (TAT): Amount of time a process spends from arrival to completion.
$$TAT = CT - AT$$
- Waiting Time (WT): Amount of time for which process waits for CPU in Ready state.
$$WT = TAT - BT$$



Topic : Scheduling Times



- Response Time (RT):





Topic : Scheduling Algorithms



1. ✓ FCFS
2. ✓ SJF
3. ✓ SRTF
4. LJF
5. LRTF
6. HRRN
7. ✓ Priority Based
8. ✓ Round Robin
9. Multilevel Queue Scheduling
10. Multilevel Feedback Queue Scheduling



Topic : FCFS (First Come First Serve)

↳ whichever process arrives first will get CPU first.

Scheduling Criteria: smaller AT first | Tie breaker $\Rightarrow$ smaller process id first

Type of Algorithm: Non-preemptive





Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time
P1	0	28
P2	0	4
P3	0	4

Gantt chart:- always starts from zero.



no. of context switches = 2



Topic : FCFS (First Come First Serve)

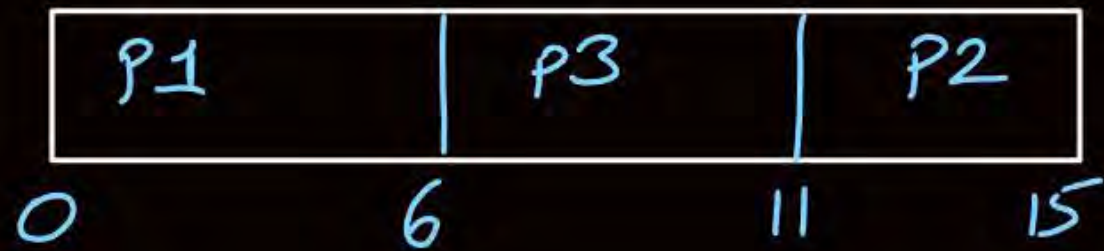
Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	28	28	28	0
P2	0	4	32	32	28
P3	0	4	36	36	32



$$\text{avg TAT} = \frac{28 + 32 + 36}{3} = 32$$

$$\text{avg WT} = \frac{0 + 28 + 32}{3} = 20$$

Process	AT	BT	CT	TAT	WT
P1	0	6	6	6	0
P2	4	4	15	11	7
P3	2	5	11	9	4



$$\text{avg TAT} = \frac{26}{3} = 8.67$$

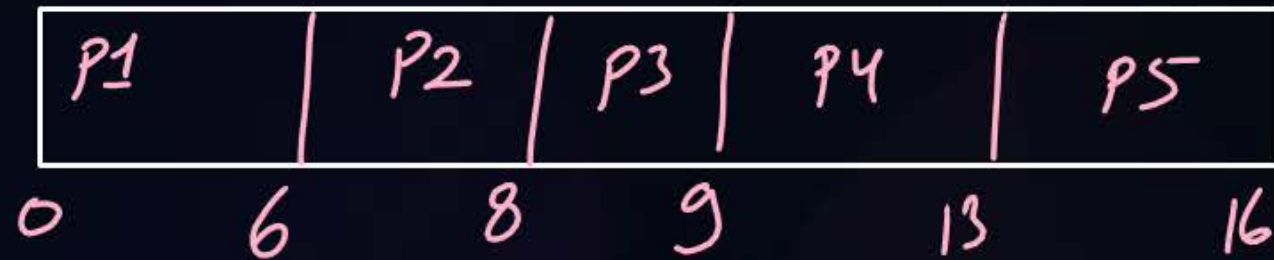
$$\text{avg WT} = \frac{11}{3} = 3.67$$



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time
P1	0	6
P2	1	2
P3	2	1
P4	3	4
P5	4	3

CT	TAT	WT
6	6	0
8	7	5
9	7	6
13	10	6
16	12	9



no. of context switches = 4

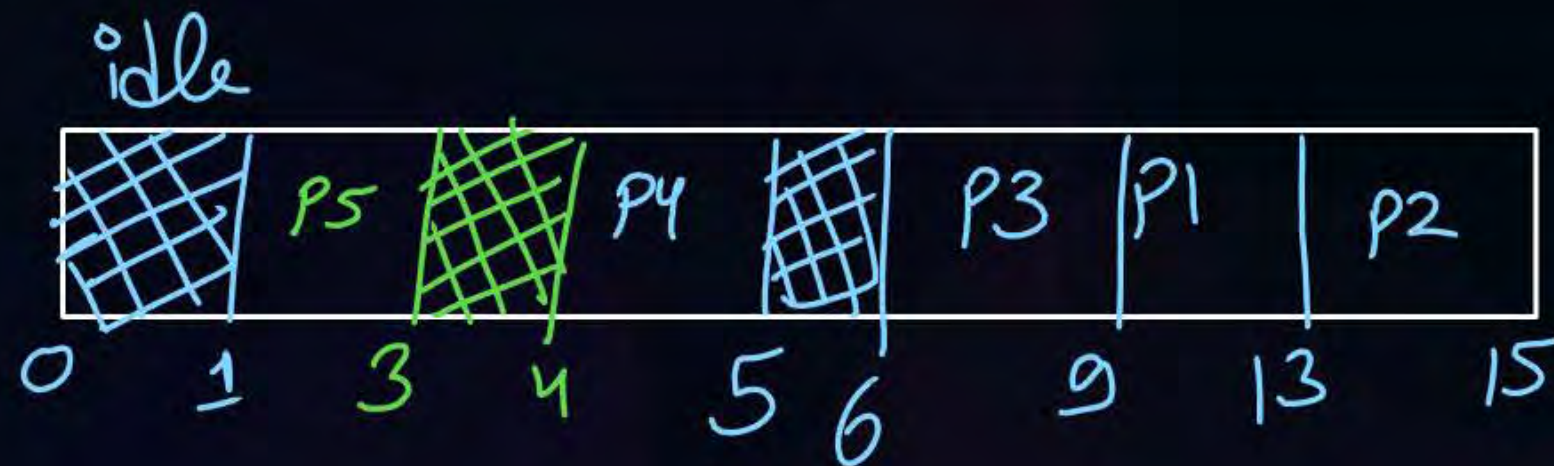
$$\text{avg TAT} = \frac{42}{5} = 8.4$$

$$\text{avg WT} = \frac{26}{5} = 5.2$$



Topic : FCFS (First Come First Serve)

Process	Arrival Time	Burst Time
P1	7	4
P2	8	2
P3	6	3
P4	4	1
P5	1	2



Total time = 15 (0 to 15)

CPU used time = 12

CPU idle time = 3

% of time CPU utilized = $\frac{12}{15} * 100\%$

—||— CPU idle = $\frac{3}{15} * 100\%$



Topic : Convoy Effect



If a long (big) process is scheduled first then execution of all other processes will be delayed.

And system will become slow.

It is known as Convoy effect.

⇒ FCFS suffers from it



Topic : FCFS (First Come First Serve)

Advantages:

1. Easy to implement
2. No complex logic
3. No starvation

Disadvantages:

1. No option of Preemption
2. Convoy effect makes the system slow





Topic : SJF (Shortest Job First)

Scheduling Criteria: *smallest BT first* | Tie breaker $\Rightarrow$ *FCFS*

Type of Algorithm: *Non-preemptive*

check BT first $\xrightarrow{\text{if equal}}$ check AT next $\xrightarrow{\text{if equal}}$ check id



Topic : SJF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	28	36	36	8
P2	0	4	4	4	0
P3	0	4	8	8	4

$$\text{avg TAT} = \frac{48}{3} = 16$$

$$\text{avg WT} = \frac{12}{3} = 4$$



H. W

Process	AT	BT
P ₁	0	4
P ₂	1	1
P ₃	2	5
P ₄	4	2



2 mins Summary

Topic

Process Scheduling

Topic

Process Scheduling Algorithms

Topic

FCFS Scheduling

Topic

SJF Scheduling



Happy Learning

THANK - YOU

